25901663 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATIONS, APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCECC102 - FUNDAMENTALS OF OS AND LINUX

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** marks

1. If a system suddenly crashes, how does the OS help recover lost files? [U] / [CO1]
2. Priority Scheduling lead to starvation.Why? [U] / [CO2]
3. A user needs to frequently access random parts of a large file. Which allocation method should be used. [U] / [CO3]
4. What way an Operating System helps the computer? [U] / [CO1]
5. Define the term for live program or acitive program. [K] / [CO2]
6. Imagine you receive a file named report.exe.Classify the file type. [U] / [CO3]
7. Which page replacement algorithm replaces the page that will not be used for the longest time in the future? [K] / [CO3]
8. What does the absence of a cycle in an resource allocation graph indicate? [U] / [CO2]
9. List the operating system services for user. [K] / [CO1]

10. Define system call. [K] / [CO1]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Describe how Time-Sharing Systems manage multiple users simultaneously. [U] / [CO1]

12. Explain the role of the PCB in process execution and management. [U] / [CO2]

13. Summarize the operations that can be performed on a directory. [U] / [CO3]

14. What are background services, and when do they run? [K] / [CO1]

15. List and define five important scheduling criteria. [K] / [CO2]

16. Why is swapping used in memory management? [K] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Describe the evolution of operating systems from early computers to modern OS. [K] / [CO1]

18. Discuss a condition that deadlock occurs.Briefly describe the deadlock characterization . [U] / [CO2]

19. Show the concept of segmentation with an example. [K] / [CO3]

[10x2 = 20]